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Estimating

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# Estimating

Estimation is a very useful math tool. As you have seen, a good estimate is a useful way to check an answer. On a multiple-choice test, a good estimate may be all that you need to choose the correct answer.

Estimating is most helpful when you can do all the work in your head. The goal in estimation is to work with “friendly” numbers. Usually, you will use rounding to make the work easier. But remember, there is no single right way to estimate.

In the examples below, three different ways of estimating are shown. When you estimate, you can check that your answer is close to your estimate.

## Example 1

Hide

3,206 × 9

Use **front-end estimation**. Round to the first digit on the left.

*Remember:* To multiply numbers that end in 0, multiply the leading digits (3 × 1 = 3). Then add on the number of zeros in the problem.

|         |        |         |
|---------|--------|---------|
| 3,206 ≈ | 3,000  | 3 zeros |
| × 9 ≈ × | 10     | 1 zero  |
| <hr/>   | <hr/>  |         |
| 28,854  | 30,000 | 4 zeros |

## Example 2

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14,340 ÷ 6

Use **compatible numbers**. Watch for relationships between numbers.

|            |          |
|------------|----------|
| 2,390      | 2,000    |
| 6)14,340 ≈ | 7)14,000 |

or

|            |          |
|------------|----------|
| 2,390      | 2,000    |
| 6)14,340 ≈ | 6)12,000 |

## Example 3

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26,379 + 24,278

Use your **knowledge of money**. If you use nickels, dimes, and quarters, you’re used to working with multiples of 5, 10, and 25. Both the numbers in this problem are close to 25,000. Because you know 25 + 25 = 50, you know the answer will be close to 50,000.

|           |         |
|-----------|---------|
| 1         | 11      |
| 26,379 ≈  | 25,000  |
| +24,278 ≈ | +25,000 |
| <hr/>     | <hr/>   |
| 50,657    | 50,000  |

*Check:* In each case, the estimate was close to the exact answer.